Lecture 8: Precise Point Positioning & Multi-GNSS

Position Determination of Ships on High seas

- Precise position necessary -> usage of GNSS
- Accuracy of navigation solution is not enough
  - Depends on Base line length
  - can become critically large
- Concepts such as VRS are not posssible due to a missing network of reference stations
- Possible solution: Precise Point Positioning (PPP)

PPP

- Aboslute position determination using a single receiver
- Similar to navigation solution
- But:
  - Usage of **dual frequency code** and carrier phase observations
  - No differencing, but accurate modelling of all error sources
- -> After sufficient measurement time the accuracy can be in the order of milimeters/centi/decimeters

Overview

- Precise satellite orbits and correction of satellite clock errors (IGS)
- Satellite antenna errors (IGS)
- Receiver antenna errors (user)
- Influence of tidal and loading effects (models)
- Influcence of atmosphere (Ionosphere, Troposphere)(user)
- Instrumental errors (user/reference station network)

Code-measurement

$$\rho_{L3}(t) = r(t) + c\left[\delta t_u(t) - \delta t^s(t-\tau)\right] + m(E)\cdot ZWD + \varepsilon_{\rho_{L3}}(t)$$

Carrier phase measurement

$$\phi_{L3}(t) = r(t) + c\left[\delta t_u(t) - \delta t^s(t-\tau)\right] + m(E)\cdot ZWD + N\cdot\lambda_{L3} + \varepsilon_{\phi_{L3}}(t)$$

$$r = \sqrt{(X - X_{SV})^2 + (Y - Y_{SV})^2 + (Z - Z_{SV})^2}$$

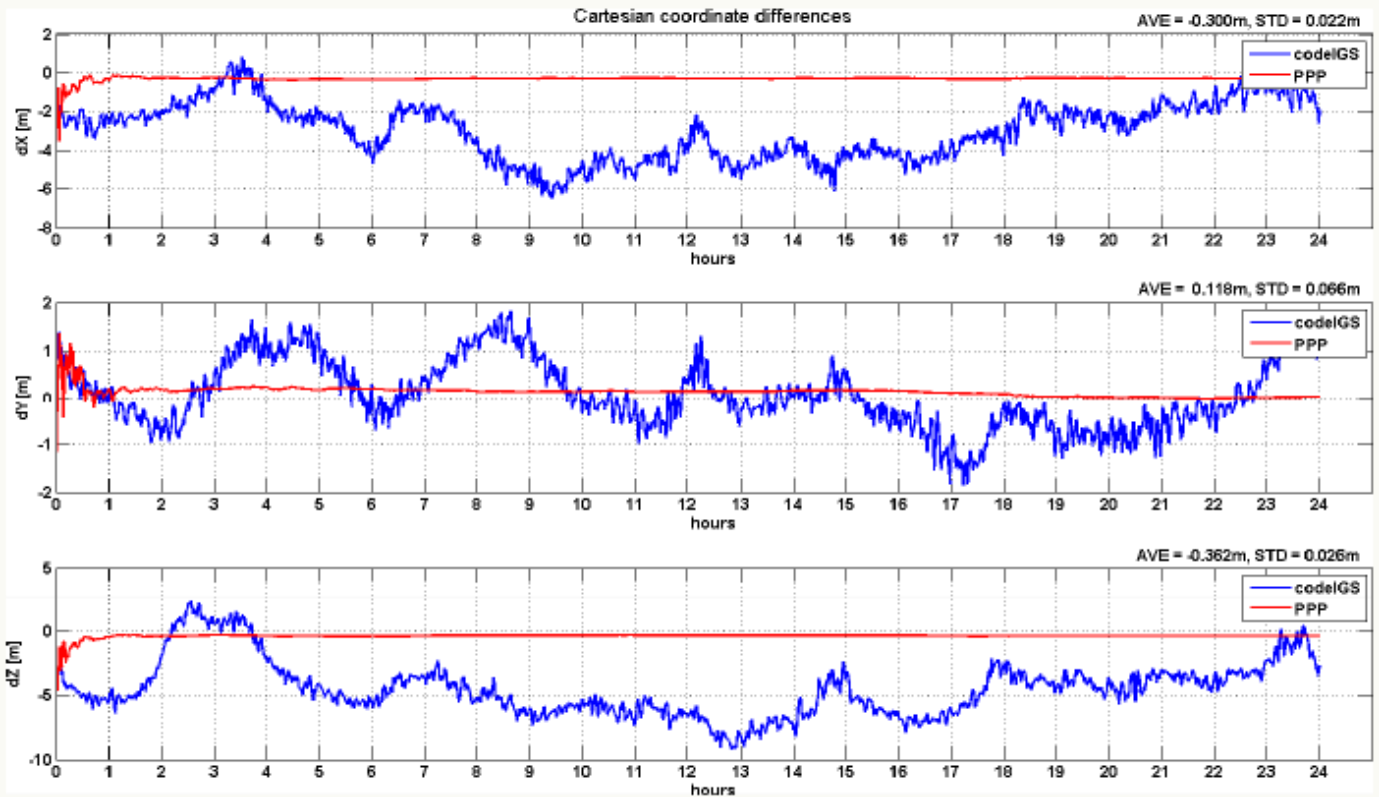
Unknown parameters in the PPP estimation :

X, Y, Z	...receiver coordinates
ZWD	...zenith wet delay (Zenith Wet Delay)
δt_u	...receiver clock error
N	...ambiguities (float)

Convergence time:
time needed until float ambiguities converge to their true values

~30 min without Augmentation

STATIC PPP- SOLUTION (GREF STATION AURICH)



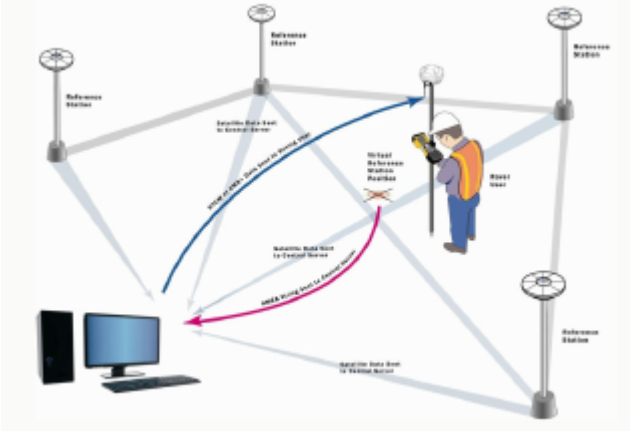
PPP is relatively new method

- not many commercial software products

- Robust and accurate processing is difficult
- Procedure:
  - Upload of observation data to the service (RINEX-Format)
  - Service calculates PPP-solution with the best possible corrections
  - Reception of the results via email or download (coordinates accuracies)

PPP-RTK

Convergence ~ 2min
Current Reference Services (e.g. SAPOS)

- Providing observations -> relative GNSS
- Modelling of errors in observation space (observation space representation - OSR)
- Lump sum is used as range correction
- Bidirectional communication
- Problem:
- -> more GNSS with additional satellites, frequencies and signals
- -> higher data bandwidth needed
- 

Idea

- Reference station networks are still used for determination of observation corrections
  - Satellite orbits and clocks
  - Atmospheric refraction
  - Etc.
- Only unidirectional communication
  - User does not transmit its position to service
- Modelling and transmission of corrections using **state space representation**
  - Not customized for individual user
- User calculates PPP solution only with own corrected observations

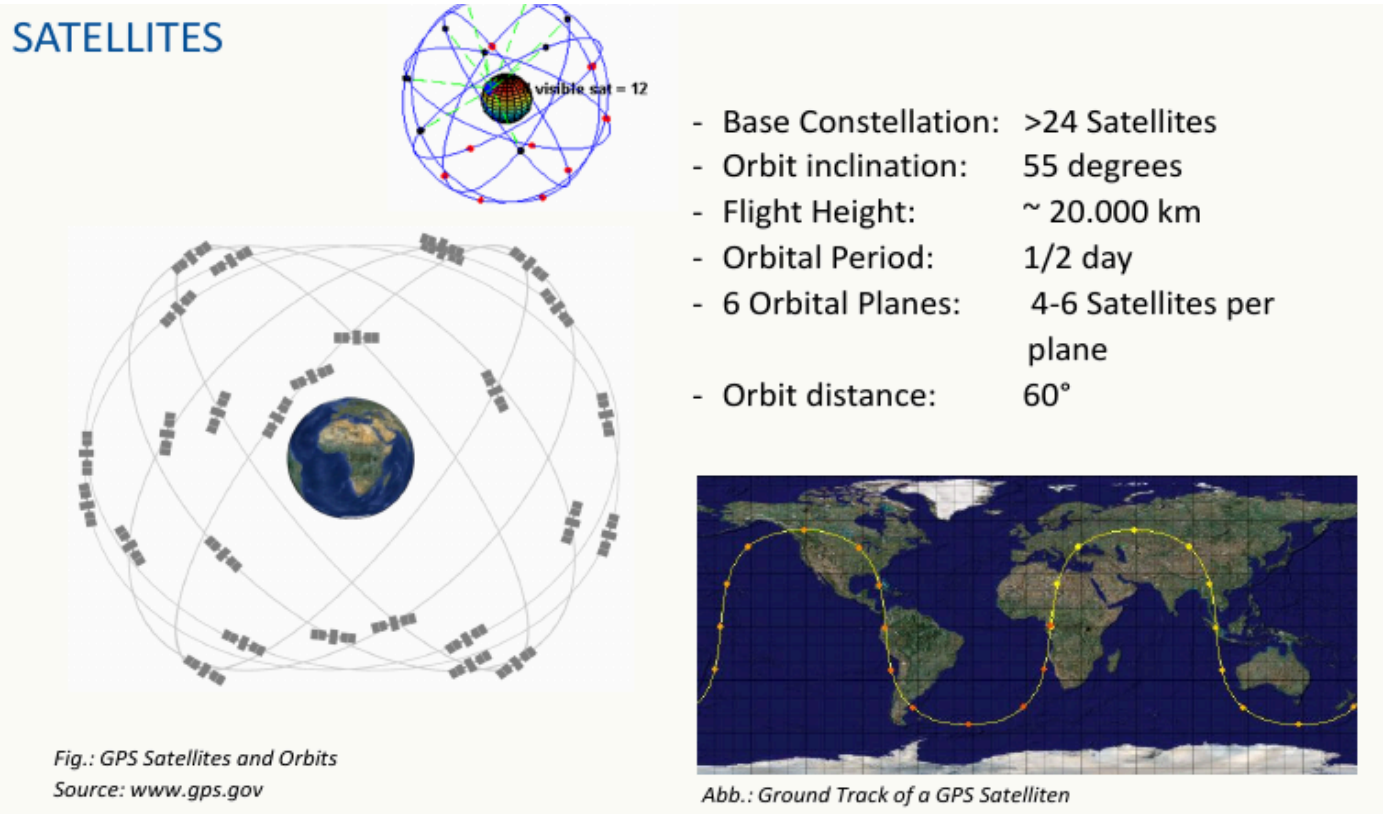
-> **Goal: centimer accurate absolute positioning in real-time**

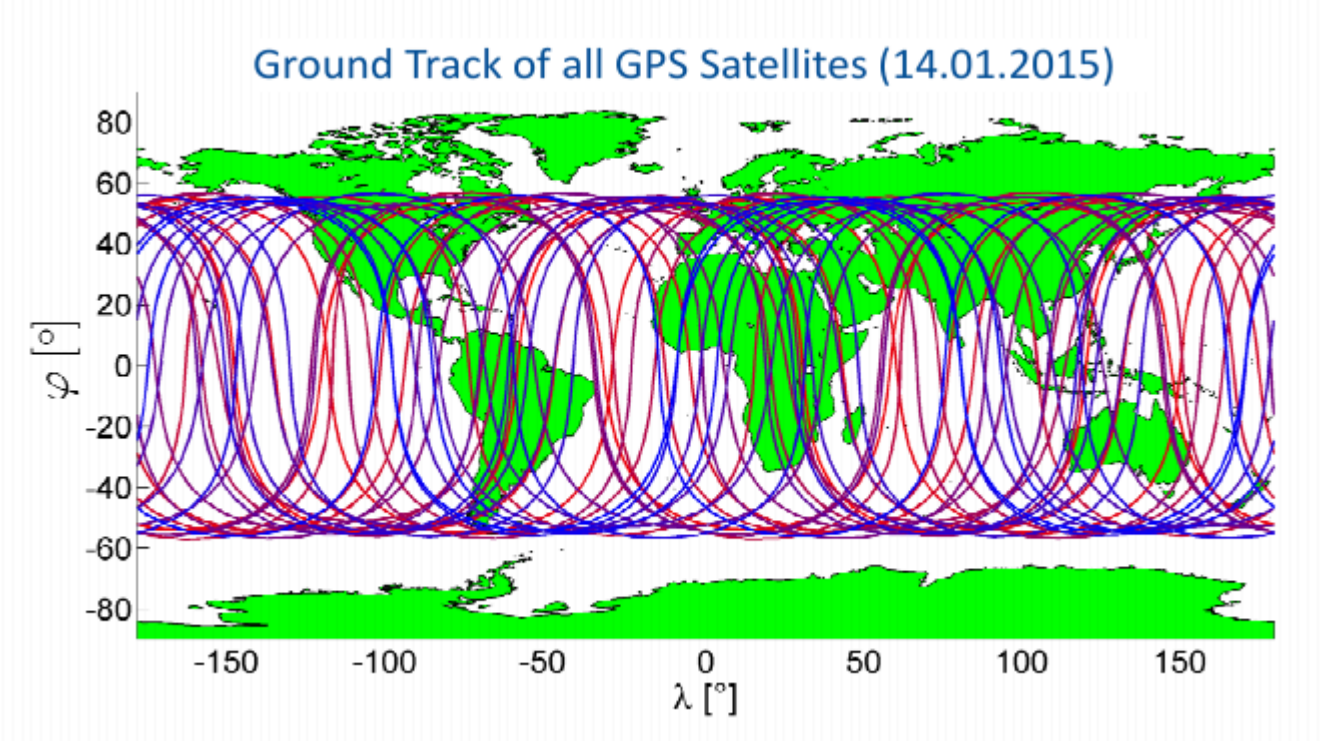
Summary

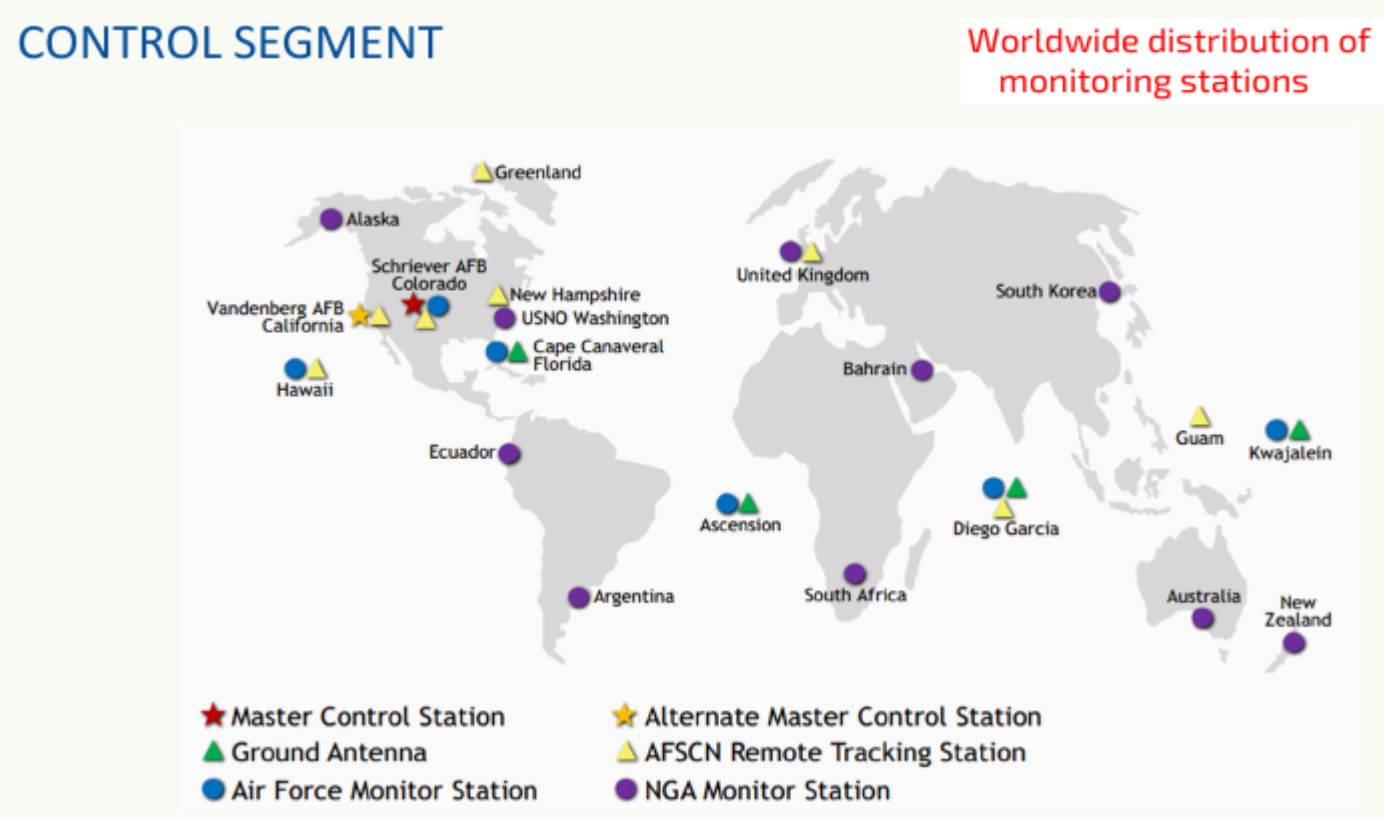
- ~dm-mm precision
- Additional IGS-products or services are necessary!
- Ambiguities are difficult to fix
- Longer convergence time for high accuracy
- Better suitable for static applications, due to long convergence times after signal interruptions
- BUT: Recently ~cm accuracy within minutes

GNSS - Global Navigation Satellite System

GPS - Global Positioning System





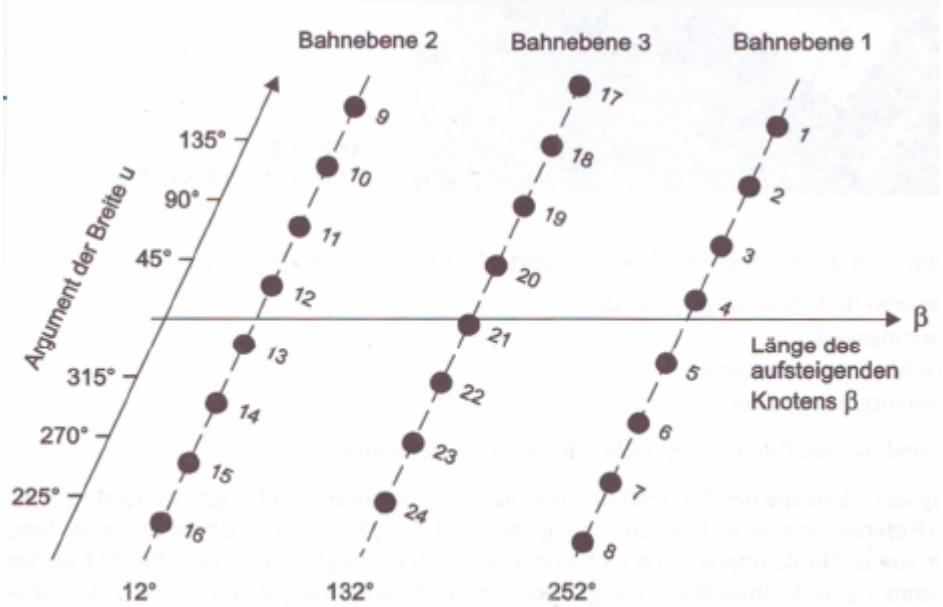


- Most commonly used GNSS (especiall mass (cheap C/A code receivers) market)
- GPS modernisation leads to significant improvements
  - Additional signals (L1C, L2C) improve SPP
  - 3rd frequency is interesting for geodetic application
    - Better ionospheric correctioin
    - Faster ambiguity fixing

- New singal structure of L1C is supposed to have ephermerides in the Nav-Message, which leads to cm-accurate orbits

GLONASS - Global'naya Navigatsionaya Sputnikovaya Sistema

Space Segments:

- 24 satellites in total
- Height: 19,100km
- Orbit inclination: 64.8° (better visibility in polar regions)
- 3 orbital planes with 8 slots
- Orbit period 11h 15min 44sec
  - -> ground track repeats every 8 days
- 

Anordnung der Satelliten in den Bahnebenen 1, 2, 3.
[1. Januar 1983 (00:00:00 Moskauer Zeit)]

Control Segments:

- Ground stations are only in Russia/SU
- Orbits and general functionality are only partially observable
- -> more difficulties in orbit calculation and frequency corrections
- -> Decreased reliability and integrity of the system

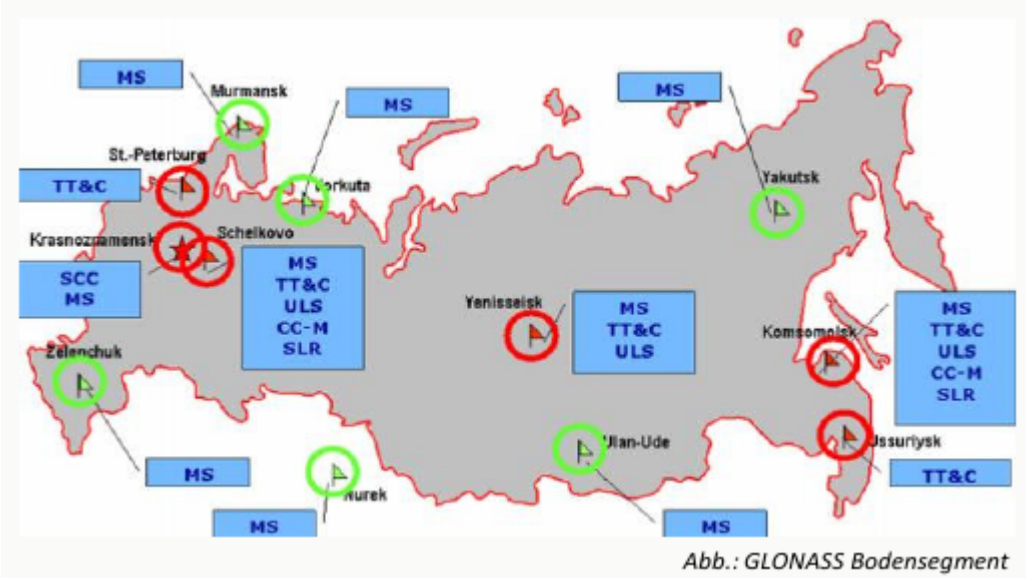
- 

Abb.: GLONASS Bodensegment

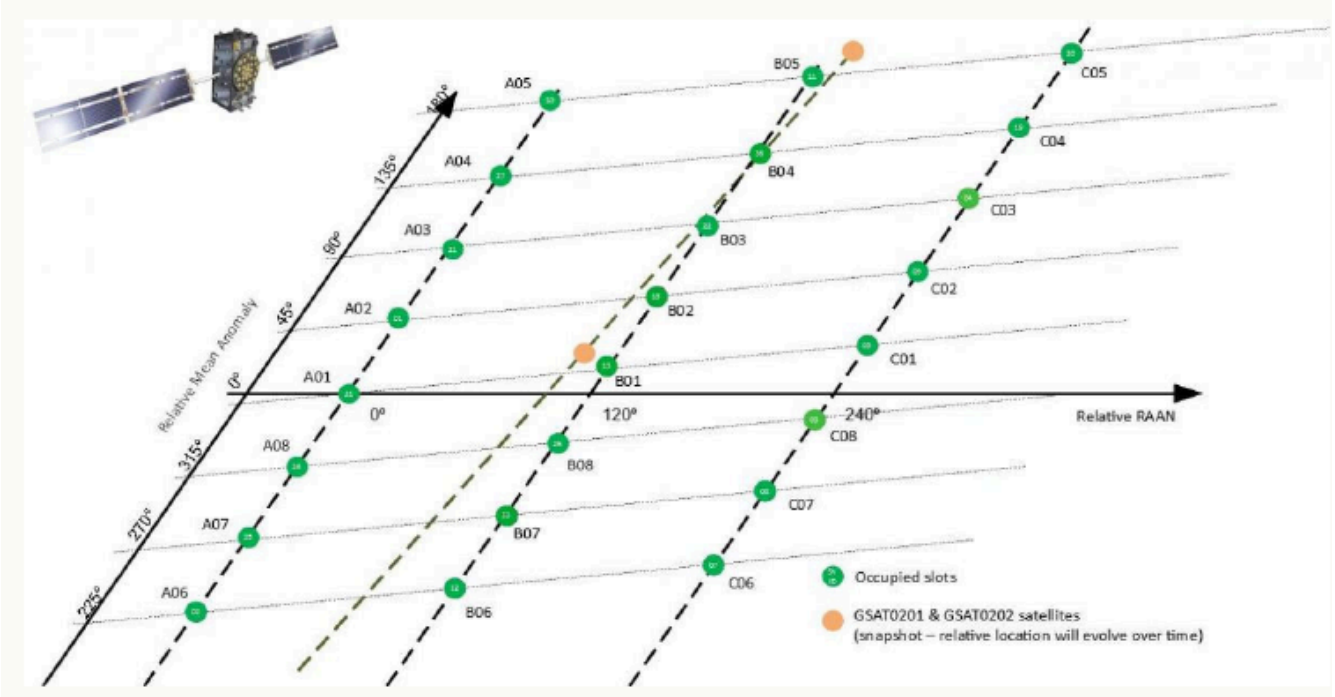
Signals (GLONASS M-Satellites, Current Constellation:

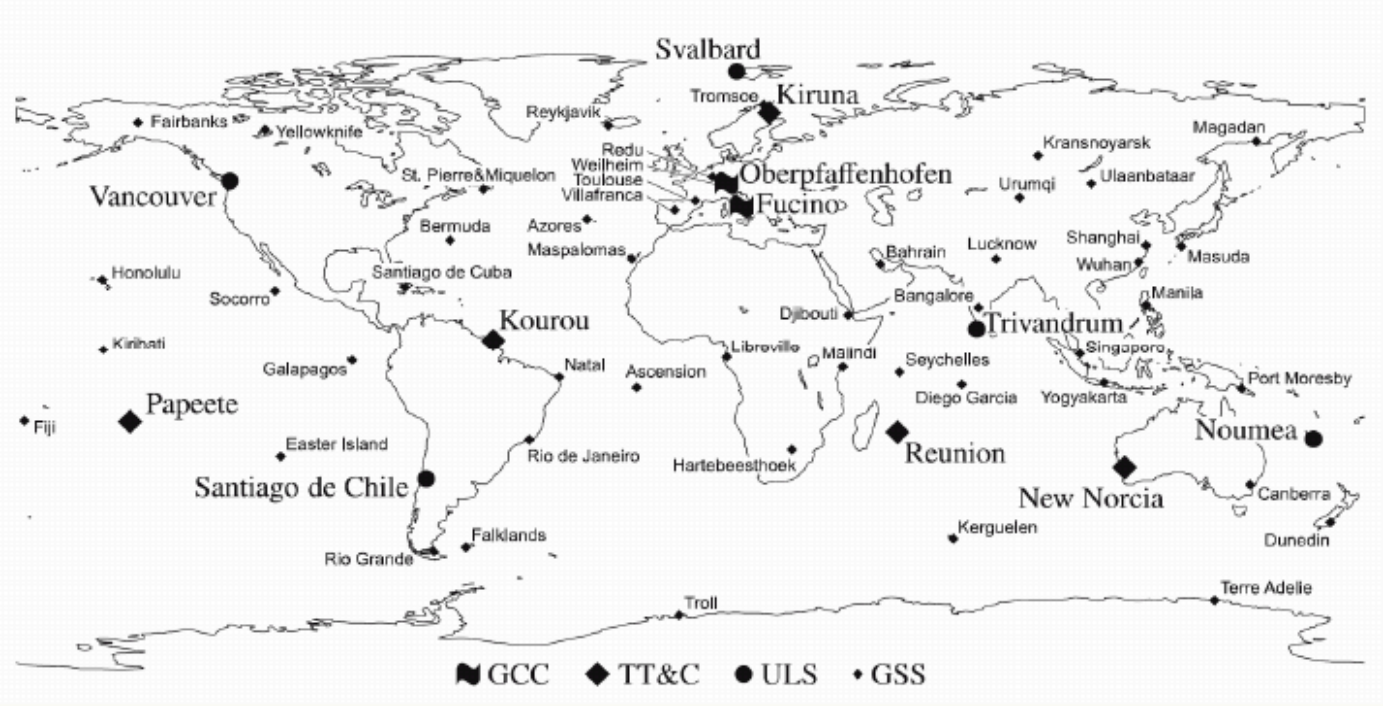
- C/A code and P-code are transmitted at 2 base frequencies
- Each satellites transmits on its own frequency (FDMA)
  -> more difficult processing (e.g. amgibuity fixing)
- Additional signals in the future (G3, G5)
- Additional availability of CDMA (Code Division Multiple Access)
- Better interoperability with other GNSS
- Reference System: Parametri Zemli
  - Earth mass/rate identical to WGS84
- Time: adapted to UTC with constant offset of 3h
- Difference to GPS time is provided by the navigation message

Summary:

- back operational for few years again (after modernisation)
- Similar accuracy as GPS
- Advantages to GPS
  - Better visibility in polar regions
  - No encryption
- Disadvantages to GPS
  - Control segments only in russia
  - Multiple frequencies increase processing complexity
- Often combined in current receivers: GPS/GLONASS

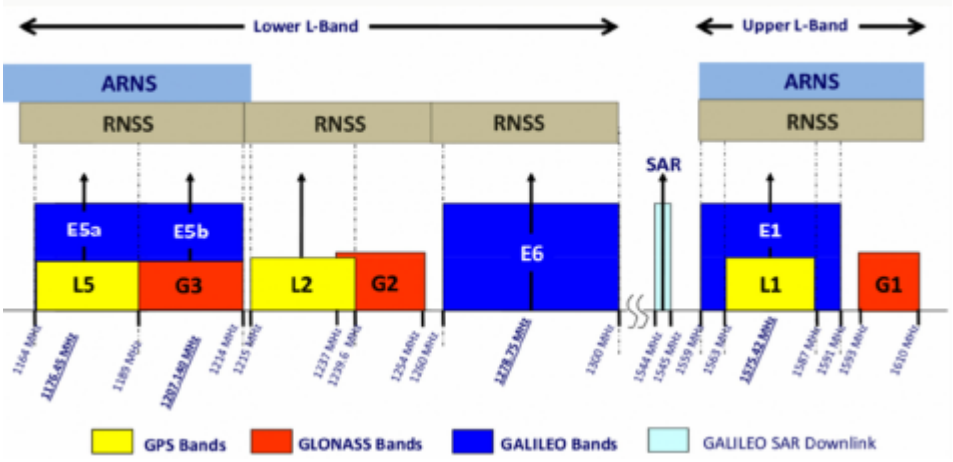
Galileo

- First GNSS planned only for civil applications
- Operated by EU and ESA (independance of member states)
- Originally planned to be operational in 2008... -> 2018
- 27 satellites in space, 25 active and usable
- Orbit inclination: 56°
- Height: 23,200km
- Orbit period: 14:04
- Ground track period: 10 days
- 3 orbital planes: 9 satellites per plane
- Orbit distance 120°
- 

- 

Signals

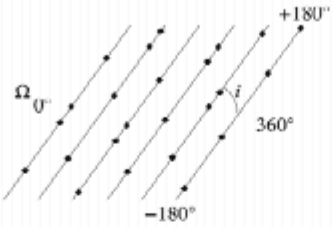
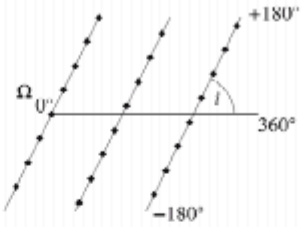
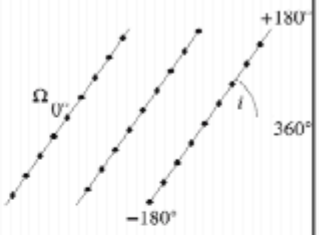
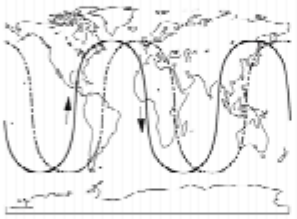
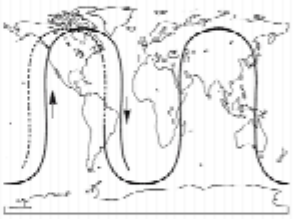
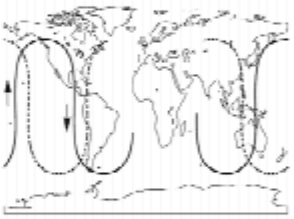
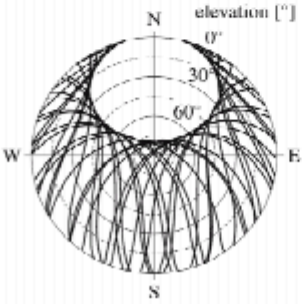
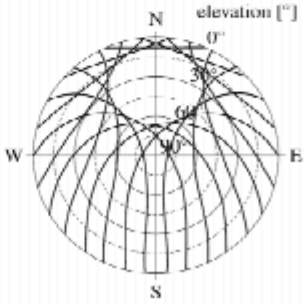
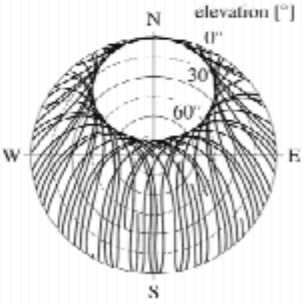
- 10 signals on 4 frequency bands -> L-Band
- C-Band as additional
- Similarity to GPS and GLONASS freqeuency



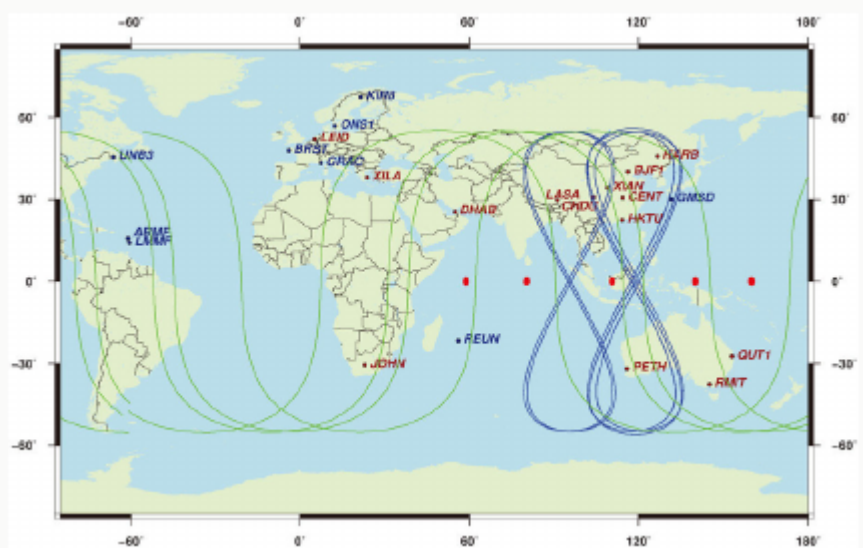
- Reference System: Galileo Terrestrial Reference Frame (GTRF)
- Independent realisation of the ITRS
- Difference between GTRF and WGS84 ~cm
  - -> Identical within the accuracy of the realisation
- Galileo System Time (GST): identical to TAI within 50ns
- Difference to GPS time provided via Nav-message

Summary:

- Strong delay in Galileo realisation
- Full constellation still a few years to go
- Interoperability to other GNSS
  - Combined procesing already possible
- Offers high potential due to modern signal structure
- and further separated frequency bands

Characteristics	GPS	GLONASS	Galileo
The constellation plot represents the inclination, the number of orbital planes, the number of satellites, and the phasing. The plots show the nominal constellation at an arbitrarily selected epoch.			
A ground track plot shows the satellite track projected onto the reference ellipsoid and in the sequel into a planar representation. The plots show 24h groundtracks of two satellites.			
A sky plot shows the azimuth and elevation of the satellites as observed by a user in a polar plot. The plots show satellites of the nominal constellation over 24 hours. User position: Graz, Austria (47.1°N, 15.5°E, 400 m)			

BEIDOU

- 35 satellites in the final constellation
  - 5 geostationary
  - 27 Medium Earth orbit satellites with 55° orbit inclination
  - 3 inclined geosynchronous satellites
  - 

- 3 frequency bands -> Overlap with Galileo bands
- Central frequencies are partial identical to central frequencies of other GLONASS
- 11 signals using CDMA

- Open Service: 10m accuracy, 20ns time accuracy
- Reference System: China Geodetic System (CGS2009)
- very similar to ITRF (~cm)
- Time: Beidou System Time (BDT), no leap seconds

Summary:

- Beidou is more and more interesting due to the increased number of MEO satellites
- System is similar to GPS and Galileo
- Very fast system realisation
- Not very well doucmented
- Possible combined processing of all other GNSS due to similar central frequencies

System	GPS	GLONASS	BeiDou	Galileo	QZSS	IRNSS/NavIC
Orbit	MEO	MEO	MEO, IGSO, GEO	MEO	IGSO, GEO	IGSO, GEO
Nominal number of satellites	24	24	27, 3, 5	30	3, 1	4, 3
Constellation	6 planes 56° inclination	Walker (24/3/1) 64.8° inclination	Walker (24/3/1) 55° inclination	Walker (24/3/1) 56° inclination	IGSOs with 43° inclination	IGSOs with 29° inclination
Services	SPS, PPS	SPS, PPS	OS, AS, WADS, SMS	OS, CS, PRS	GCS, GAS, PRS, EWS, MCS	SPS, RS
Initial service	Dec 1993	Sep 1993	Dec 2012	2016/2017 (planned)	2018 (planned)	2016 (planned)
Origin	USA	Russia	China	Europe	Japan	India
Coverage	Global	Global	Global	Global	East Asia Oceania region	$-30° < \phi < 50°$ $30° < \lambda < 130°$
Frequency (MHz)	L1 1575.42 L2 1227.60 L5 1176.45	L1 1602.00 L2 1246.00 L3 1202.025	B1 1561.098 B2 1207.14 B3 1268.52	E1 1575.42 E5a 1176.45 E5b 1207.14 E6 1278.75	L1 1575.42 L2 1227.60 L5 1176.45 E6 1278.75	L5 1176.45 S 2492.028